INCH-POUND

MIL-C-24643/54C
14 March 1994
SUPERSEDING
MIL-C-24643/54B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS8NW6
(INCLUDING VARIATION LS8NWA6)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS8NW6 (unarmored) and LS8NWA6 (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, AWG size 12, class B stranding.

* Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Cross-linked polyethylene or ethylene propylene rubber insulation. Minimum average thickness 0.030 inch. Standard identification code shall be applied by method 1 or method 3.

Fourth - Eight conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well-rounded assembly.

Fifth - Binder tape applied helically with overlap.

AMSC N/A FSC 6145

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Construction (non-watertight) (continued)

Construction variant for type LS8NW6

* Sixth - Cross-linked polyolefin jacket. Minimum average thickness 0.040 inch. Maximum overall diameter 0.670 inch.

* Seventh- Braided metal armor.

Notes: 1. Variant LS8NW6 military part number - M24643/54-01UN.
2. Variant LS8NW6A military part number - M24643/54-02AN

EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	1.72
Voltage withstand - volts, root mean square, minimum	4000
* Insulation resistance - Conductor to conductor, Megohms/1000 feet, minimum	4200
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200 °C)	50
Cold bending, cable - 5 inch mandrel diameter maximum	No damage
Tear - pounds per inch thickness, minimum ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80

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EXAMINATION AND TESTS:	Requirements
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) cycles, minimum	25
* Permanence of printing (jacket) - (type LS8NW6 only) cycles, minimum (method 1 only)	125
* Armor - Type LS8NWA6 only, conformance to material, construction and coverage	No failure

* Group D:

Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 +/- 2°C)	No damage
Aging and compatibility (cable) (125 +/- 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 2000 feet nominal

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NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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